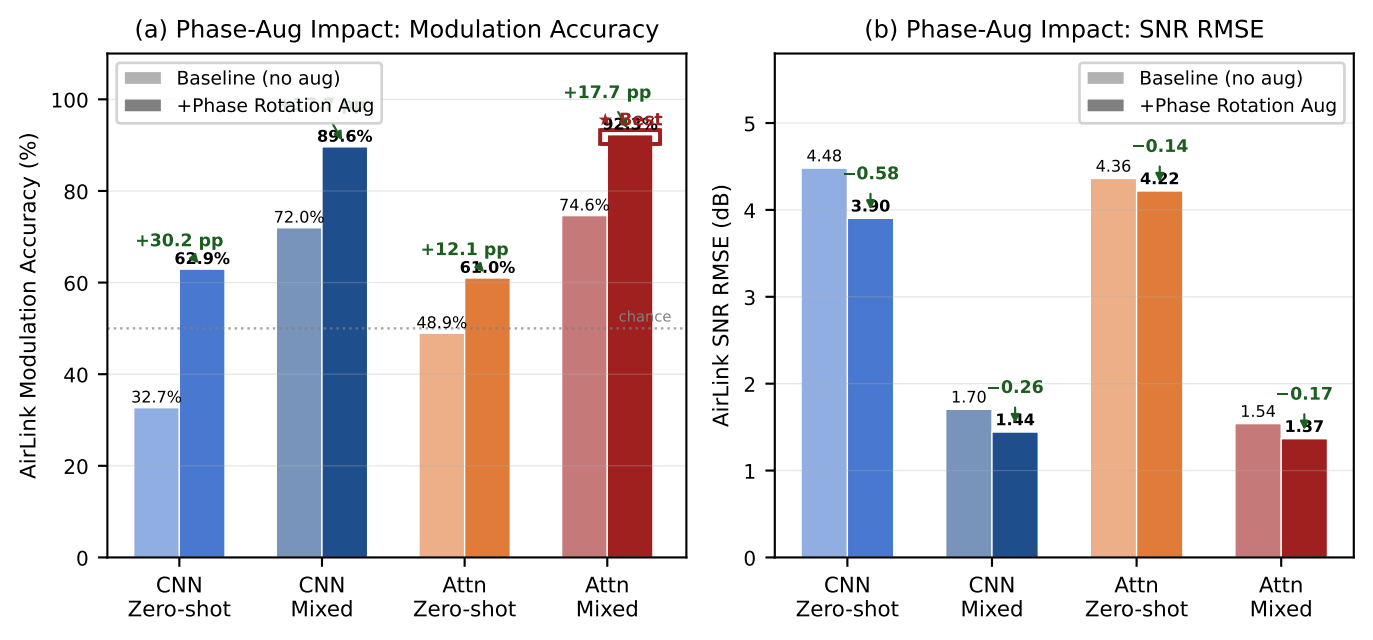
(a) Phase-Aug Impact: Modulation Accuracy
Baseline (no aug)
+Phase Rotation Aug
AirLink Modulation Accuracy (%)
100
80
60
40
20
0
+30.2 pp
62.9%
32.7%
72.0%
89.6%
+12.1 pp
61.0%
48.9%
+17.7 pp
★ Best
92.3%
74.6%
chance
CNN Zero-shot
CNN Mixed
Attn Zero-shot
Attn Mixed
(b) Phase-Aug Impact: SNR RMSE
Baseline (no aug)
+Phase Rotation Aug
AirLink SNR RMSE (dB)
5
4
3
2
1
0
4.48
−0.58
3.90
1.70
−0.26
1.44
4.36
−0.14
4.22
1.54
−0.17
1.37
CNN Zero-shot
CNN Mixed
Attn Zero-shot
Attn Mixed